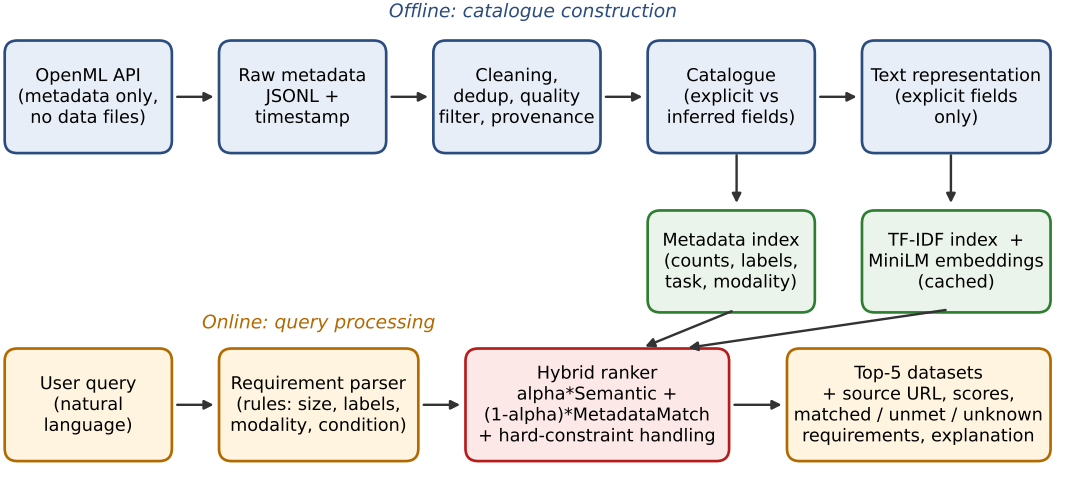
Offline: catalogue construction
OpenML API (metadata only, no data files)
Raw metadata JSONL + timestamp
Cleaning, dedup, quality filter, provenance
Catalogue (explicit vs inferred fields)
Text representation (explicit fields only)
Metadata index (counts, labels, task, modality)
TF-IDF index + MiniLM embeddings (cached)
Online: query processing
User query (natural language)
Requirement parser (rules: size, labels, modality, condition)
Hybrid ranker alpha*Semantic + (1-alpha)*MetadataMatch + hard-constraint handling
Top-5 datasets + source URL, scores, matched / unmet / unknown requirements, explanation